

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Simulate linear convolution and circular convolution on discrete time signals.	
Experiment No: 2	Date: 18/07/2026	Enrolment No:92400133075

GITHUB : <https://github.com/Chiragsinh933/DSIP.git>

AIM: Simulate linear convolution and circular convolution on discrete time signals.

CODE :

```
import numpy as np
import matplotlib.pyplot as plt

def linear_convolution(signal1, signal2):
    linear_conv = np.convolve(signal1, signal2, mode='full')
    return linear_conv

def circular_convolution(signal1, signal2):
    fft_length = len(signal1) + len(signal2) - 1
    fft_signal1 = np.fft.fft(signal1, fft_length)
    fft_signal2 = np.fft.fft(signal2, fft_length)
    circular_conv = np.fft.ifft(fft_signal1 * fft_signal2)
    return circular_conv

signal1 = np.array([3, 5, 1, 2, 4])
signal2 = np.array([2, 4, 6, 8, 10])

linear_conv = linear_convolution(signal1, signal2)
circular_conv = circular_convolution(signal1, signal2)

plt.figure(figsize=(10, 6))

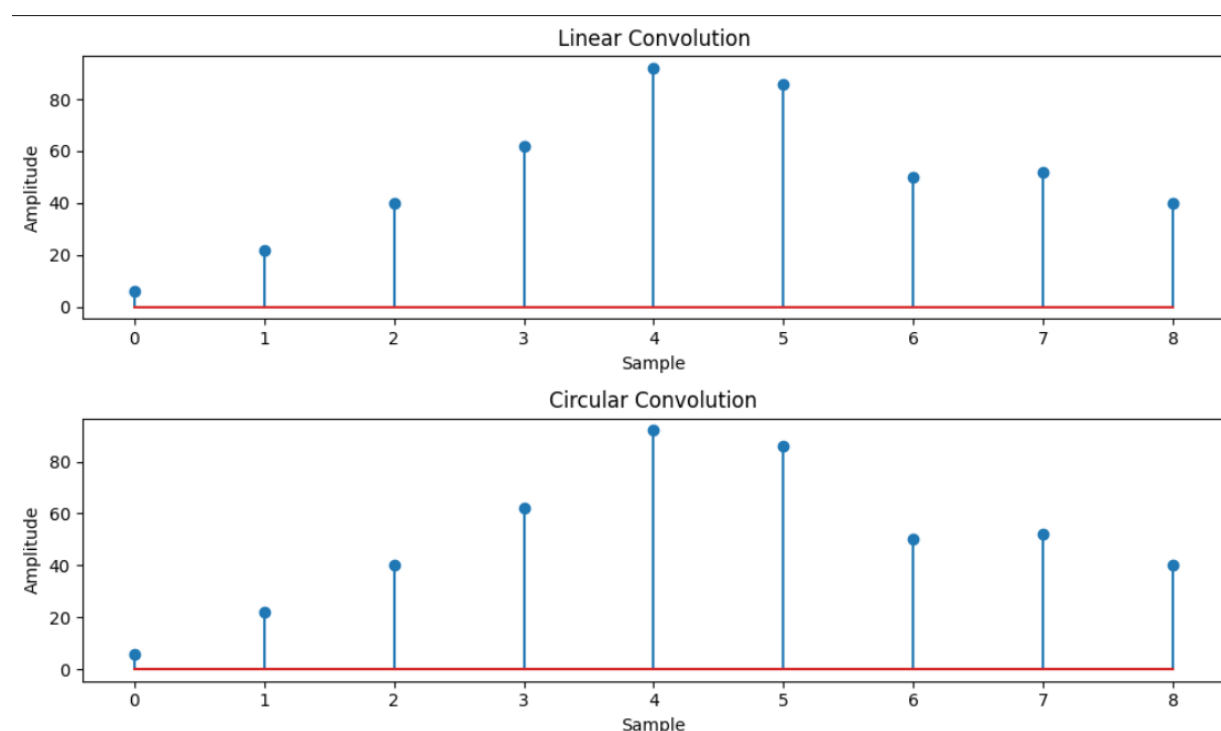
plt.subplot(2, 1, 1)
plt.stem(linear_conv)
plt.title('Linear Convolution')
plt.xlabel('Sample')
plt.ylabel('Amplitude')

plt.subplot(2, 1, 2)
plt.stem(np.real(circular_conv))
plt.title('Circular Convolution')
plt.xlabel('Sample')
plt.ylabel('Amplitude')

plt.tight_layout()
plt.show()
```

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OUTPUT :



Conclusion :

In this experiment We learned how to perform **linear convolution** and **circular convolution** on discrete-time signals using Python. The output demonstrated that linear convolution produces the complete response of two signals, while circular convolution performs convolution under periodic (cyclic) conditions using the FFT method.

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Calculation :



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Experiment - 2.

Q. linear convolution.

signal 1 = $[3, 5, 1, 2, 4]$

signal 2 = $[2, 4, 6, 8, 10]$

3	5	1	2	4	
2	6	10	2	4	8
4	12	20	4	8	16
6	18	30	6	12	24
8	24	40	8	16	32
10	30	50	10	20	40

$$y(0) = 6$$

$$y(1) = 10 + 12 = 22$$

$$y(2) = 2 + 20 + 18 = 40$$

$$y(3) = 4 + 4 + 30 + 24 = 62$$

$$y(4) = 8 + 8 + 6 + 40 + 30 = 92$$

$$y(5) = 16 + 12 + 8 + 50 = 86$$

$$y(6) = 24 + 16 + 10 = 50$$

$$y(7) = 32 + 20 = 52$$

$$y(8) = 40$$

$$y[n] = [6, 22, 40, 62, 92, 86, 50, 52, 40]$$

⇒ circular convolution.

$$N = 5 + 5 - 1 = 9$$

$$x_1 = [3, 5, 1, 2, 4, 0, 0, 0, 0]$$

$$x_2 = [2, 4, 6, 8, 10, 0, 0, 0, 0]$$



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$y(0)$ 3 0 0 0 0 4 2 1 5
 $y(1)$ 3 5 0 0 0 0 4 2 1
 $y(2)$ 1 5 3 0 0 0 0 4 2
 $y(3)$ 2 1 5 3 0 0 0 0 4
 $y(4)$ 4 2 1 5 3 0 0 0 0
 $y(5)$ 0 4 2 1 5 3 0 0 0
 $y(6)$ 0 0 4 2 1 5 3 0 0
 $y(7)$ 0 0 0 4 2 1 5 3 0

$y(0)$	3	5	1	2	4	0	0	0	0	2
$y(1)$	0	3	5	1	2	4	0	0	0	4
$y(2)$	0	0	3	5	1	2	4	0	0	6
$y(3)$	0	0	0	3	5	1	2	4	0	8
$y(4)$	0	0	0	0	3	5	1	2	4	10
$y(5)$	0	4	0	0	0	3	5	1	2	0
$y(6)$	2	4	0	0	0	0	3	5	1	0
$y(7)$	1	2	4	0	0	0	0	3	5	0
$y(8)$	5	1	2	4	0	0	0	0	3	0

$$y(0) = (3 \times 2) = 6$$

$$y(1) = (5 \times 2) + (3 \times 4) = 10 + 12 = 22$$

$$y(2) = (1 \times 2) + (5 \times 4) + (3 \times 6) = 40$$

$$y(3) = (2 \times 2) + (1 \times 4) + (5 \times 6) + (3 \times 8) = 62$$

$$y(4) = (4 \times 2) + (2 \times 4) + (1 \times 6) + (5 \times 8) + (3 \times 10) = 92$$

$$y(5) = (0 \times 2) + (4 \times 4) + (2 \times 6) + (1 \times 8) + (5 \times 10) = 86$$

$$y(6) = (0 \times 2) + (0 \times 4) + (4 \times 6) + (2 \times 8) + (1 \times 10) = 50$$

$$y(7) = (0 \times 2) + (0 \times 4) + (0 \times 6) + (4 \times 8) + (2 \times 10) = 52$$

$$y(8) = (0 \times 2) + (0 \times 4) + (0 \times 6) + (0 \times 8) + (4 \times 10) = 40$$

$$y(n) = [6, 22, 40, 62, 92, 86, 50, 52, 40]$$

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